

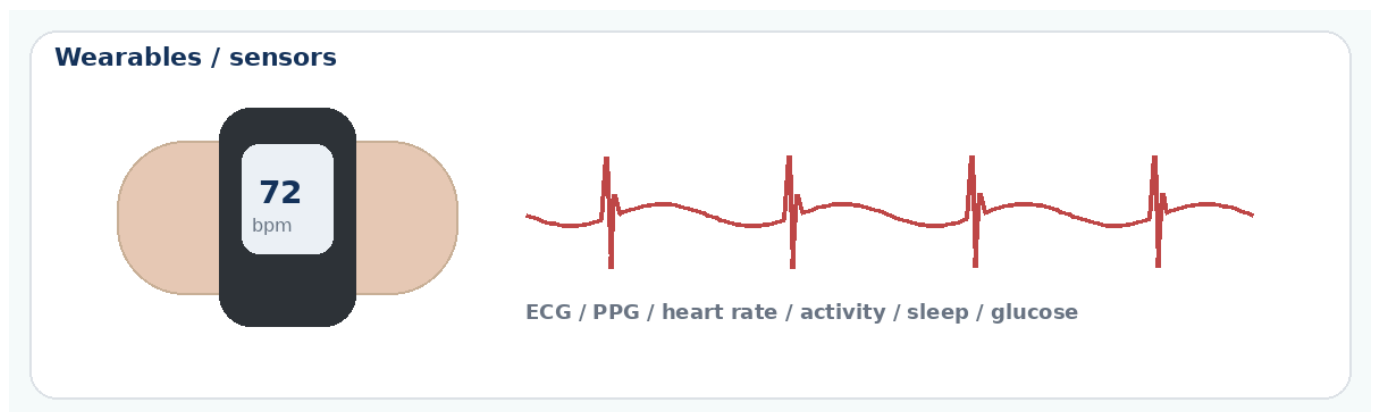
Biomedical & Public Health Data Types

A compact AI research guide with representative data, search keywords, frequent model families, and a visual example for each modality.

1. Wearables / Sensors

Continuous or repeated measurements collected from wearable devices and physiological sensors. They are especially useful for remote monitoring, digital biomarkers, and time-series prediction.

Typical data / samples	Key search keywords	Frequent models
ECG, PPG, smartwatch, accelerometer, heart rate, SpO ₂ , sleep, activity, continuous glucose monitoring	wearable sensing; digital biomarkers; physiological time series; remote patient monitoring; activity recognition; sleep staging	1D-CNN; LSTM/GRU; TCN; Transformers; Random Forest; XGBoost

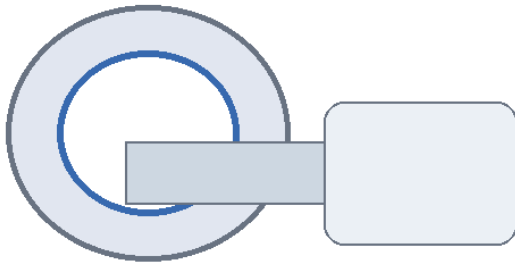


Representative schematic illustration

2. Medical Imaging

Images acquired with clinical imaging systems to visualize anatomy, physiology, lesions, or disease progression. Common AI tasks include classification, detection, segmentation, and prognosis.

Typical data / samples	Key search keywords	Frequent models
CT, MRI, X-ray, ultrasound, PET, mammography, retinal imaging, radiology images	medical image analysis; radiology AI; image segmentation; lesion detection; radiomics; computer-aided diagnosis	ResNet; DenseNet; EfficientNet; U-Net/nnU-Net; ViT; Swin Transformer; DETR

Medical imaging

CT • MRI • X-ray • PET • ultrasound

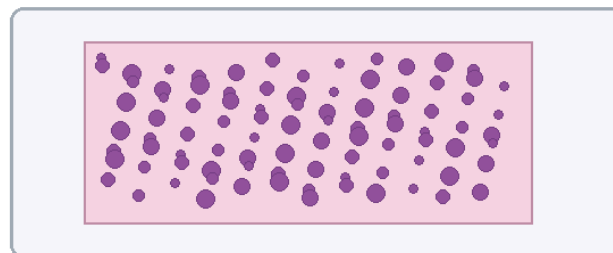
Representative schematic illustration**3. Histopathology**

Microscopic analysis of tissue specimens, usually obtained through biopsy or surgery. Digital pathology converts stained tissue slides into images that can be analyzed at patch or whole-slide level.

Typical data / samples	Key search keywords	Frequent models
biopsy, tissue sample, H&E slide, immunohistochemistry (IHC), pathology slide, whole-slide image (WSI), tumor tissue	biopsy; histology; computational pathology; digital pathology; WSI; H&E; IHC; cancer grading; multiple instance learning	ResNet/EfficientNet; ViT/Swin; MIL; Attention-MIL; CLAM; self-supervised learning; pathology foundation models

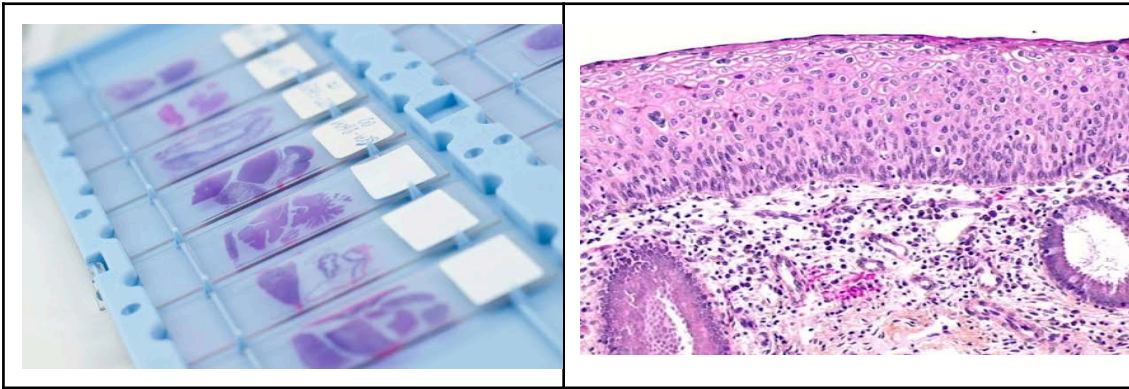
Histopathology

Biopsy / tissue sample



H&E / IHC / whole-slide image (WSI)

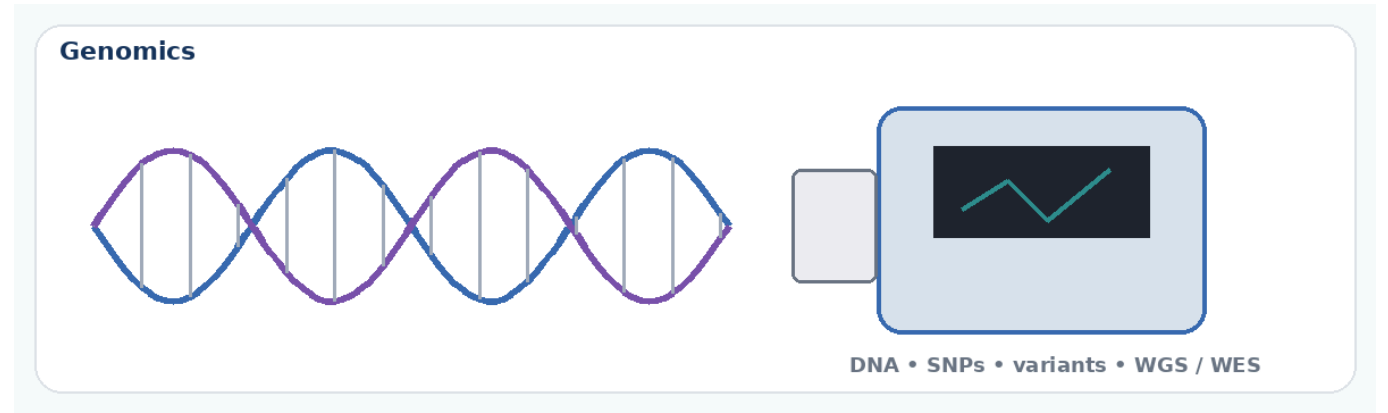
Representative schematic illustration



4. Genomics

Data describing DNA sequence and genetic variation. AI is used for variant interpretation, regulatory prediction, phenotype association, and sequence-based modeling.

Typical data / samples	Key search keywords	Frequent models
DNA, SNPs, mutations, genetic variants, WGS, WES, sequencing reads, genotype matrices	genomic prediction; GWAS; variant calling; whole-genome sequencing; sequence modeling; regulatory genomics	Logistic/linear models; LASSO/Elastic Net; Random Forest; XGBoost; CNNs; DNA Transformers



Representative schematic illustration

5. Transcriptomics

Measurements of RNA abundance that describe which genes are being expressed in a sample. These data are widely used for disease subtyping, biomarkers, pathways, and treatment-response studies.

Typical data / samples	Key search keywords	Frequent models
RNA, RNA-seq, mRNA, gene-expression matrix, bulk RNA-seq, transcript counts	gene expression; differential expression; RNA-seq; transcriptomic signatures; disease subtyping; pathway analysis	Elastic Net; Random Forest; XGBoost; autoencoders/VAEs; GNNs; Transformers

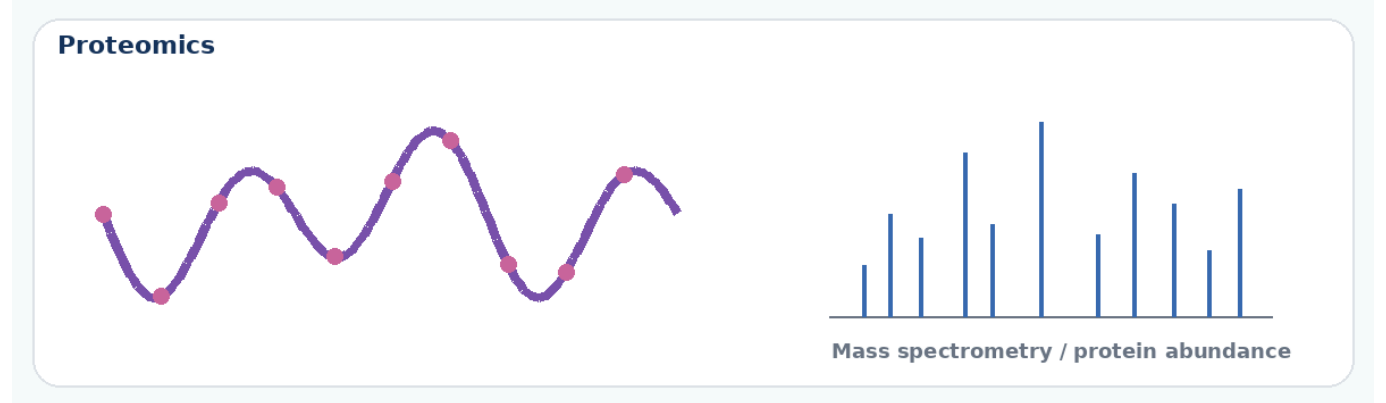


Representative schematic illustration

6. Proteomics

Large-scale measurements of proteins, peptides, and protein abundance. Proteomics connects molecular function more directly to phenotype and is frequently generated using mass spectrometry.

Typical data / samples	Key search keywords	Frequent models
proteins, peptides, protein abundance, mass spectrometry, LC-MS/MS, protein expression, biomarkers	proteomics; protein identification; protein abundance; mass spectrometry; protein biomarkers; protein function	SVM; Random Forest; XGBoost; neural networks; GNNs; protein language models

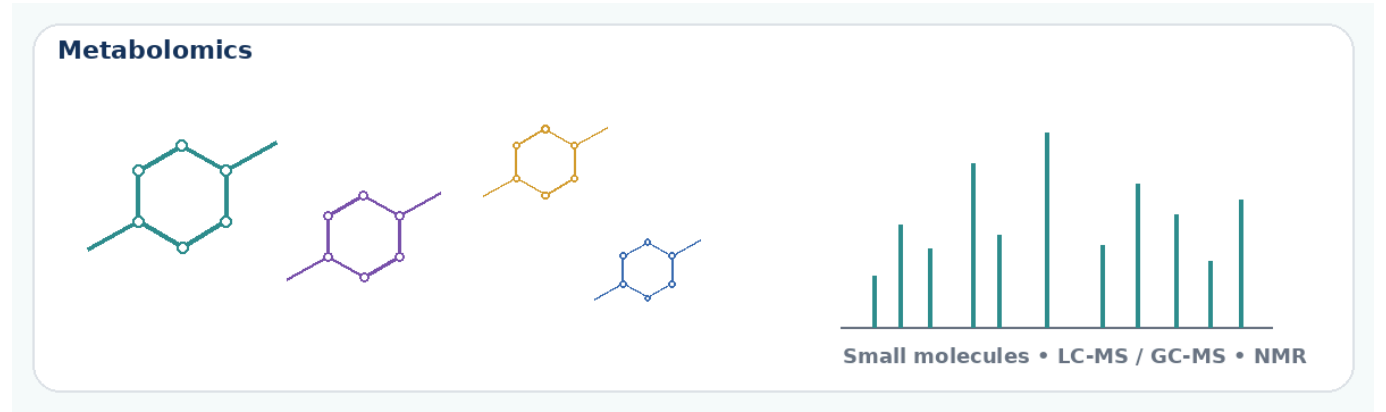


Representative schematic illustration

7. Metabolomics

Measurements of small molecules and metabolites in biological samples such as blood, plasma, serum, or urine. They provide a snapshot of biochemical state and metabolic pathways.

Typical data / samples	Key search keywords	Frequent models
metabolites, serum, plasma, urine, LC-MS, GC-MS, NMR, metabolic profiles	metabolomics; metabolite profiling; metabolic biomarkers; untargeted metabolomics; pathway analysis	PLS-DA; SVM; Random Forest; XGBoost; Elastic Net; autoencoders; neural networks

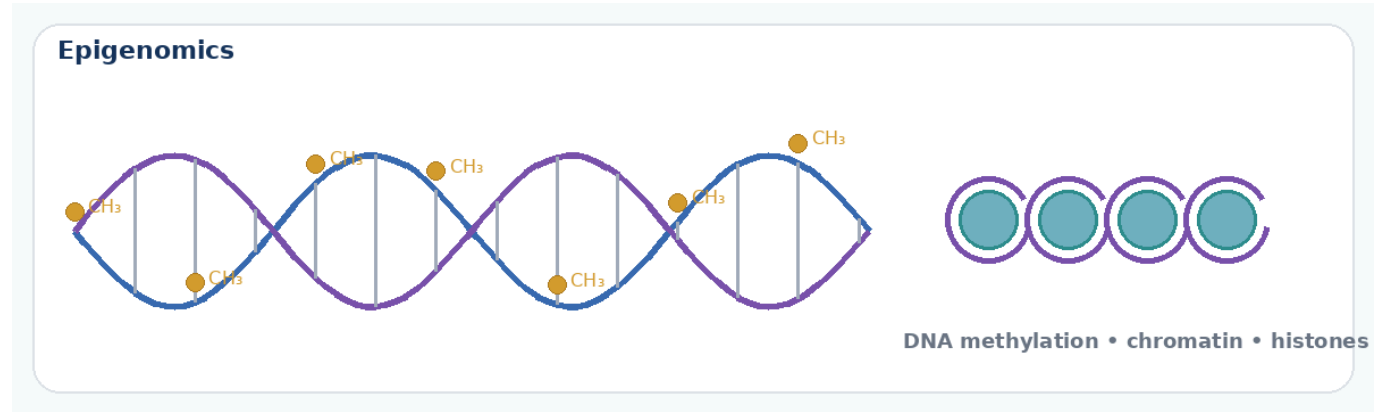


Representative schematic illustration

8. Epigenomics

Data describing regulatory modifications that affect gene activity without changing DNA sequence. Common measurements capture DNA methylation, chromatin accessibility, and histone marks.

Typical data / samples	Key search keywords	Frequent models
DNA methylation, methylation arrays, ATAC-seq, ChIP-seq, chromatin, histone modifications	epigenomics; DNA methylation; chromatin accessibility; ATAC-seq; ChIP-seq; epigenetic biomarkers	Elastic Net; Random Forest; XGBoost; CNNs; Transformers; autoencoders

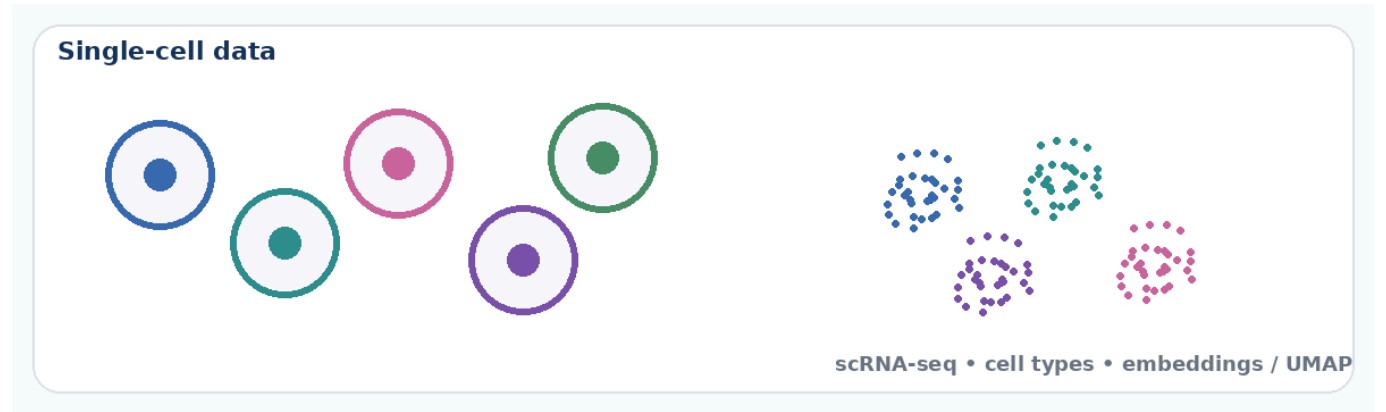


Representative schematic illustration

9. Single-cell Data

Measurements collected at individual-cell resolution, allowing researchers to study cellular heterogeneity, rare cell populations, developmental trajectories, and disease-specific cell states.

Typical data / samples	Key search keywords	Frequent models
scRNA-seq, single cells, cell-by-gene matrix, cell type, cell state, cell atlas, spatial single-cell data	single-cell RNA-seq; cell type annotation; clustering; trajectory inference; cell-cell communication; cell atlas	PCA/UMAP + clustering; scVI/VAEs; GNNs; Transformers; scGPT-like foundation models

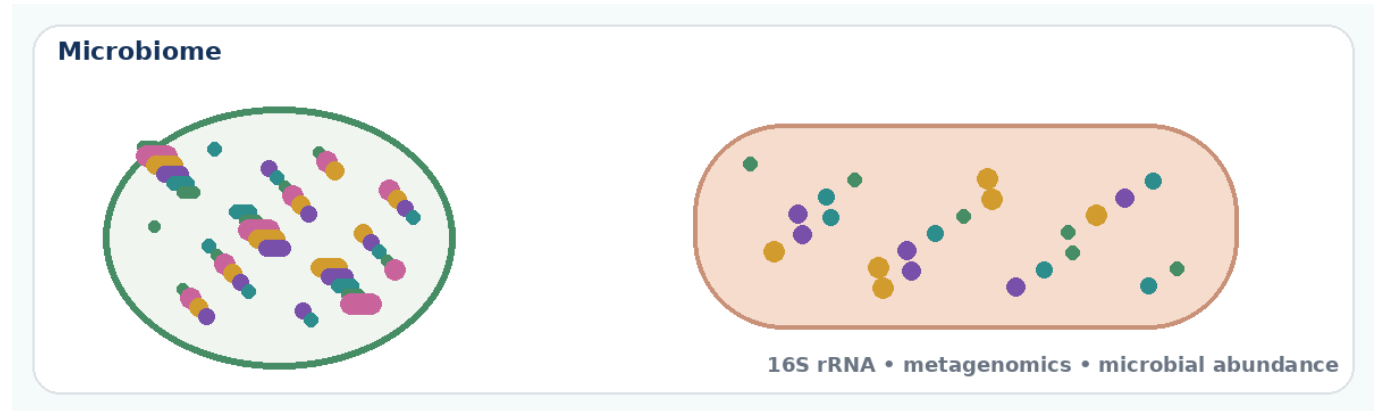


Representative schematic illustration

10. Microbiome

Data describing microbial communities living in the gut, mouth, skin, or other environments. AI can relate community composition and microbial functions to disease, diet, or treatment response.

Typical data / samples	Key search keywords	Frequent models
stool/fecal sample, gut microbiome, oral microbiome, 16S rRNA, metagenomics, microbial abundance	microbiome; metagenomics; 16S rRNA; microbial community; dysbiosis; microbiome-disease association	Random Forest; XGBoost; SVM; compositional models; GNNs; Transformers



Representative schematic illustration

11. Clinical Trials

Prospective intervention data collected under a predefined protocol, often with treatment and control groups. Key analyses concern efficacy, safety, survival, heterogeneity, and causal treatment effects.

Typical data / samples	Key search keywords	Frequent models
participants, treatment arm, placebo/control, intervention, outcomes, adverse events, follow-up, survival time	randomized controlled trial; treatment response; survival analysis; causal inference; patient stratification; treatment effect	Cox models; Random Survival Forest; DeepSurv; causal forests; Bayesian models; uplift models

Clinical trials



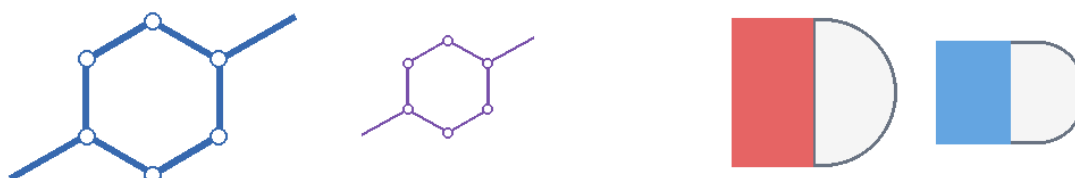
Representative schematic illustration

12. Drug / Molecular Data

Information about compounds, molecular structures, targets, binding, pharmacology, and toxicity. It underpins computational drug discovery, repurposing, and molecular property prediction.

Typical data / samples	Key search keywords	Frequent models
molecules, compounds, SMILES, molecular graphs, drug targets, binding affinity, ADMET, assays	drug discovery; QSAR; molecular property prediction; drug-target interaction; molecular docking; ADMET; drug repurposing	Random Forest; XGBoost; GNN/MPNN; Graph Transformers; molecular language models; VAEs/diffusion models

Drug / molecular data



SMILES • structure • target • ADMET • binding

Representative schematic illustration

13. Environmental / Exposure Data

Measurements of external exposures and environmental conditions that influence health. These data are often spatial and temporal and can be linked to population or individual health outcomes

Typical data / samples	Key search keywords	Frequent models
air pollution, PM2.5, temperature, rainfall, water quality, satellite data, land use, vectors, geospatial exposure	environmental epidemiology; climate-health; exposure modeling; geospatial health; air pollution; vector-borne disease	GAMs/spatial models; Random Forest; XGBoost; LSTM; Transformers; GNNs; spatiotemporal models

Environmental / exposure data



Air pollution / PM2.5



Climate • satellite • geospatial exposure

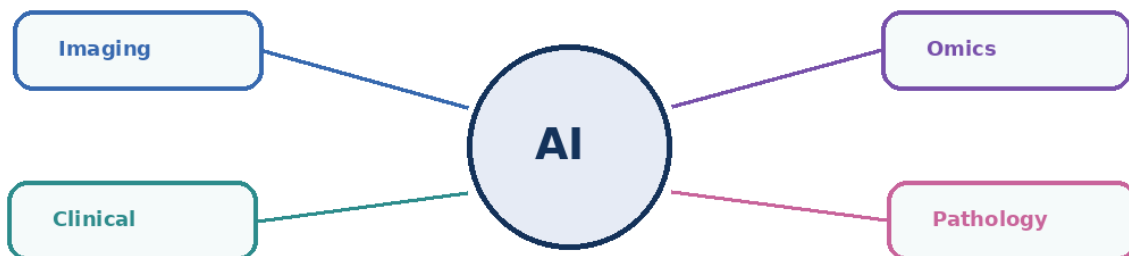
Representative schematic illustration

14. Multimodal Biomedical Data

Joint analysis of two or more data modalities, such as imaging, pathology, clinical variables, genomics, or other omics. The goal is to learn complementary information that is unavailable from a single modality.

Typical data / samples	Key search keywords	Frequent models
imaging + clinical, pathology + genomics, radiology + omics, EHR + biomarkers, multi-omics, longitudinal multimodal data	multimodal learning; data fusion; radiogenomics; image-genomics integration; multi-omics integration; multimodal foundation models	Early/late fusion; multimodal Transformers; cross-attention; contrastive learning; GNNs; VAEs

Multimodal biomedical data



Fusion • cross-attention • contrastive learning

Representative schematic illustration